

Problem Set for 20 March 2025

In [LADR4e](#),

1. $\mathcal{L}(U, V) := \text{Hom}_{\mathbb{F}}(U, V)$, and $\mathcal{L}(U) := \text{End}_{\mathbb{F}}(U)$ as the ground field $\mathbb{F}$ is fixed;
2. $\text{null} = \ker$, and $\text{range} = \text{im}$;

Problems from [LADR4e](#).

1. (3A-17) Suppose V is finite-dimensional. Show that the only two-sided ideals of $\mathcal{L}(V)$ are $\{0\}$ and $\mathcal{L}(V)$.
 - A subspace $\mathcal{E}$ of $\mathcal{L}(V)$ is called a two-sided ideal of $\mathcal{L}(V)$ if $TE \in \mathcal{E}$ and $ET \in \mathcal{E}$ for all $E \in \mathcal{E}$ and all $T \in \mathcal{L}(V)$.
 - **(optional)** Find a non-trivial two-sided ideal of $\mathcal{L}(\mathbb{F}[x])$.
2. (3B-22) Suppose U and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \text{null } ST \leq \dim \text{null } S + \dim \text{null } T.$$

- **(optional)** Both U and V can be ∞ -dimensional. The inequality holds as long as $\dim \text{null } S + \dim \text{null } T < \infty$.
3. (3B-23) Suppose U and V are finite-dimensional vector spaces and $S \in \mathcal{L}(V, W)$ and $T \in \mathcal{L}(U, V)$. Prove that

$$\dim \text{range } ST \leq \min\{\dim \text{range } S, \dim \text{range } T\}.$$

- **(optional)** Both U and V can be ∞ -dimensional. The inequality holds as long as $\dim \text{range } S < \infty$ or $\dim \text{range } T < \infty$.
4. (3D-17) Suppose V is finite-dimensional and $S \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by $\mathcal{A}(T) = ST$ for $T \in \mathcal{L}(V)$.
 1. Show that $\dim \text{null } \mathcal{A} = (\dim V)(\dim \text{null } S)$.
 2. Show that $\dim \text{range } \mathcal{A} = (\dim V)(\dim \text{range } S)$.
 3. **(optional)** Write down the matrix form of $\mathcal{A}$ under the basis $\{v_{i \cdot n + j} := E_{i,j}\}_{1 \leq i,j \leq n}$.

Since there are some [solutions](#) on the website, it is recommended to finish all optional problems.